### 9.6.7. Green Revolution

Green revolution the term was coined by William S.Gaud in (1968). It is defined as the cumulative result of a series of research, development, innovation and technology transfer initiatives. Agricultural production (especially wheat and rice) manifolds worldwide particularly in the developing countries between the 1940's and the late 1960's.

The Green revolution or third Agricultural Revolution is the intensive plan of 1960's to increase crop yield in developing countries by introducing the high yielding, resistant varieties, increased irrigation facilities, fertilizer application and better agricultural management.

In 1963 semi-dwarf wheat of Mexico was introduced from which India got five prolonged strategies for breeding a wide range of high varieties like Sonora 64, Sonalika and Kalyansona possessing a broad spectrum of resistance to major biotic and abiotic condition. Same as wheat M.S.Swaminathan produced the first semi-dwarf fertiliser responsive hybrid variety of rice_TNI (Taichung Native-1) in 1956 from Taiwan. The derivatives were introduced in 1966. Later better yielding semi dwarf varieties of rice Jaya and Ratna developed in India.




NORIN 10 – The cultivars found that Norin 10 dwarfing genes have high photosynthetic rate per unit leaf area and increase respiratory activity. Gonjiro Inazuka selected the semi-dwarf wheat variety that became Norin 10. He would have never thought that the semi dwarf genes would not only revolutionize the world of wheat but also helped to save more than one billion lives from hunger and starvation.

### Plant Breeding for Developing Resistance to diseases

Some crop varieties bred by hybridization and selection, for disease resistance to fungi, bacteria and viral diseases are released (Table 9.1).

| Crop | Variety | Resistance to diseases |
|---|---|---|
| Wheat | Himgiri | Leaf and Stripe rust, hill bunt |
| Brassica | Pusa swarnim (Kara rai) | White rust |
| Cauliflower | Pusa Shubhra, Pusa snowball K-1 | Black rot and curl blight black rot |
| Cowpea | Pusa Komal | Bacterial blight |
| Chilli | Pusa Sadabahar | Chilly mosaic virus, Tobacco mosaic virus and Leaf curl. |

**Table 9.2** Disease resistance varieties

Resistance to yellow mosaic virus in bhindi *(Abelmoschus escullentus)* was transferred from a wild species and resulted in a new variety of *A. Escullentus* called Parbharni kranti.

### Plant Breeding for Developing Resistance to Insect Pests

Insect resistance in host crop plants may be due to morphological, biochemical or physiological characteristics. Hairy leaves in several plants are associated with resistance to insect pests. Example: resistance to jassids in cotton and cereal leaf beetle in wheat. In wheat, solid stems lead to non-preference by the stem sawfly and smooth leaves and nectar-less cotton varieties do not attract bollworms. High aspartic acid, low nitrogen and sugar content in maize leads to resistance to maize stem borers.

**Sugar cane:** *Saccharum bareri* was originally grown in North India, but had poor sugar content and yield. Tropical canes grown in South India *Saccharum officinarum* had thicker stems and higher sugar content but did not grow well in North India. These two species were successfully crossed to get sugar cane varieties combining the desirable qualities of high yield, thick stems, high sugar and ability to grow in the sugarcane areas of North India.



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